

MUHAMMAD HAIDER ZAIDI

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🌐 github.com/chefhaider 🏠 Bavaria, Germany 🎂 21.06.2000

Technical Skills

Programming: Python, C++

Tools/Frameworks: Docker, FastAPI, AWS SageMaker, Kubernetes, TensorFlow, PyTorch, YOLO

Competence: Deep Learning, Machine Learning, MLOps, Computer Vision, Medical Image Analysis, Computational Geometry, 3D Data Processing, Algorithm Optimization, Numerical Stability, Rigorous Validation, Signal Processing

Experience

German National Metrology Institute (PTB)

March 2024 – Present

Student Assistant Researcher

Berlin, DE

- Designed and evaluated **deep learning models** for **metrological applications**, enhancing **model performance** and **training efficiency**.
- Explored **novel architectures** and **physics-informed hybrid loss functions** to improve **model accuracy** and **stability**.
- Collaborated with PhD researchers on a publication for the FLOMEKO conference, contributing to **research and exploration of new methods**.
- Built **dashboards** for **data visualisation** and **model performance monitoring** for **deep learning models**.

Folio3

March 2022 – Nov 2023

Machine Learning Engineer

California, US (remote)

- Improved vitiligo **segmentation accuracy** to 94% and increased speed by 5x using **AI algorithms**; deployed with **Docker** and **FastAPI**.
- Developed a **real-time** tennis foul detection system with **TensorRT**, reducing **compute cost** and time by 6x, demonstrating **algorithm optimization**.
- Managed **MLOps pipelines** on **AWS SageMaker**, proactively addressing **YOLO model data drift**, and maintained **Kubernetes infrastructure**.
- Developed **Proof-of-Concepts (POCs)** using **YOLO**, **few-shot learning**, and **CNN** for client demonstrations, translating clinical needs into **algorithm designs**.

Manda GmbH

March 2024 - June 2024

Werkstudent Python Developer

Nürnberg, DE

- Built an **AI chatbot application** using the **OpenAI API** and relevant **LLM tools/frameworks**, demonstrating **deep learning algorithm** application.
- Contributed to full-stack development of a website providing **AI-based services** using **Flask**, **HTML**, and **CSS**.

Freelance

July 2020 – March 2022

Data Engineer

Remote

- Built **web scrapers** and **ETL pipelines** for structured and unstructured data, ensuring robust data foundations for **AI/ML applications**.
- Automated **data ingestion and cleaning workflows** for client analytics, improving data quality and efficiency.

Projects

CAFA 6: Protein Function Prediction

2022

Deep Learning Competition

- Architected a **weighted-probability ensemble** integrating DPFunc and DeepGOPlus, utilizing **AlphaFold** for **feature extraction** and **taxonomic bias mitigation**.
- Increased **model accuracy** by 4% through custom **feature engineering** across MF, CC, and BP namespaces for **protein function prediction**.

Razanlytics: Real-time Bitcoin Price Forecasting

2022

Thesis Project

- Developed a thesis project focusing on **real-time Bitcoin price forecasting** using **deep learning models** and **time-series analysis**.

Saaz: AI Self-Playing Chitralli Sitar

2023

Signal Processing, Music Generation

- Collaborated to build the world's first **AI-driven self-playing Chitralli Sitar**, combining **signal processing** and **music generation LLMs**.

Hugging Face

Open-Source

Contributor

- Co-authored the **Zero-shot Image Classification technical documentation and implementation page** on the Hugging Face Hub.

Education

Friedrich Alexander Universität

Present

Master of Science in Artificial Intelligence

Erlangen, DE

- Currently pursuing a **Master of Science in Artificial Intelligence**, specializing in **deep learning** and **algorithm design**.

National University of Computer and Emerging Sciences

Aug 2018 – June 2022

Bachelor of Science in Computer Science

Karachi, PK

- Completed **Bachelor of Science in Computer Science**, with coursework in **data structures**, **algorithms**, and **machine learning**.
- Awarded **Dean's List of Honor** for academic excellence.

Certifications

PyTorch: Deep Neural Networks with PyTorch | **Coursera:** TensorFlow Developer Certificate
| **Databricks:** Lakehouse Fundamentals | **Datacamp:** Computer Vision in Python